# EL MEHDI HARESS

Researcher in stochastic modelling | PhD in Applied Mathematics | CentraleSupélec engineer

Dynamic models | Estimation from imperfect data | Uncertainty and robustness | Python

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Sophia Antipolis, France — open to relocating to Paris

## PROFILE

Researcher with **5 years** of work on **how a state evolves over time** and how to learn it from data: stochastic dynamical models, **estimation of several parameters from discrete, noisy observations**, convergence and **stability guarantees**, and **risk-aware learning** robust to heavy tails (2 *AISTATS* papers). Eight peer-reviewed papers; I have driven my own research from the question to the published result, and released the code. I want to bring this to world models of patient health, where calibration and robustness decide whether a model can be trusted in the clinic.

## RESEARCH STRENGTHS

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| **Temporal & dynamic** | Stochastic differential equations, long-memory processes (fractional Brownian motion), long-time behaviour and stability, multi-scale systems, simulation with proven error bounds. |
| **Uncertainty** | Parameter inference with convergence rates, sensitivity of a model to misspecification, Monte Carlo, spectral risk measures (CVaR), heavy-tailed and unbounded losses. |
| **Engineering** | **Python** (NumPy, SciPy, pandas) — 8 years; Git; MATLAB, R; TypeScript, React, Firebase. Open-source research code (5 public repositories). Neural network basics. |
| **Communication** | 15+ international talks (invited: Warwick, IHP, Rennes); reviewer for journals; teaching from undergraduate to M2. English — fluent (2 years in the UK); French — native. |

## EXPERIENCE

09/2026 — now
**Research engineer — probabilistic numerical methods**
Inria Sophia Antipolis — CALISTO team (Mireille Bossy)
- Lagrangian stochastic models for turbulence and multi-scale industrial systems; work with engineers and industrial partners. Python implementation and numerical validation.

12/2024 — 08/2026
**Research fellow in stochastic analysis**
University of Leeds (UK) — K. Dareiotis and K. Lê
- Uniform pathwise stability of singular dynamics with memory (*SPA*, 2026): how solutions react to changes in the model and its parameters.
- Co-organised FRSSD 2026 (18 speakers). Weekly research collaboration with a PhD student.

10/2021 — 11/2024
**PhD — numerical approximation of singular SDEs and SPDEs**
Université Paris-Saclay / CentraleSupélec — L. Goudenège and A. Richard
- Simulation schemes with proven convergence where the standard assumptions fail (*IMA J. Numerical Analysis*, *SPA*).
- **Joint estimators of drift, noise level and memory** from discretely observed trajectories, with convergence rates (*Statistical Inference for Stochastic Processes*); implemented and released.
- Taught probability, PDEs and stochastic finance (M2) at CentraleSupélec.

03/2020 — 09/2021
**Research intern — learning under risk constraints**
Osaka University (Japan) — M. Holland
- Learning algorithms that optimise spectral risk instead of the mean, with guarantees under unbounded losses and heavy tails; 2 *AISTATS* papers with implementations and experiments.

09/2019 — 02/2020
**Research intern — inference for long-memory processes**
University of Alberta (Canada) — Y. Hu
- Simultaneous estimation of all parameters of a fractional Ornstein–Uhlenbeck model from discrete data.

## EDUCATION

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| **2021--2024** | PhD in Applied Mathematics — Université Paris-Saclay. |
| **2017--2021** | Engineering degree, Applied Mathematics — CentraleSupélec. M2 Mathematics of Randomness (2020–2021). |
| **2015--2017** | CPGE (MP) — Lycée Pierre de Fermat, Toulouse. |

## SELECTED PUBLICATIONS

8 papers in total; full list at elmehdiharess.org. Author order is alphabetical in probability.

- M. Holland, **E.M. Haress**. *Spectral risk-based learning using unbounded losses*. AISTATS, 2022.
- M. Holland, **E.M. Haress**. *Learning with risk-averse feedback under potentially heavy tails*. AISTATS, 2021.
- **E.M. Haress**, A. Richard. *Estimation of several parameters in discretely-observed SDEs with additive fractional noise*. Statistical Inference for Stochastic Processes, 2024.
- K. Dareiotis, **E.M. Haress**, K. Lê. *Uniform pathwise stability of additive singular SDEs driven by fractional Brownian motion*. Stochastic Processes and their Applications, 2026.
- L. Goudenège, **E.M. Haress**, A. Richard. *Numerical approximation of SDEs with distributional drift*. Stochastic Processes and their Applications, 2025.